

Ensemble spaces: a common foundation for probability, information theory and physical theories

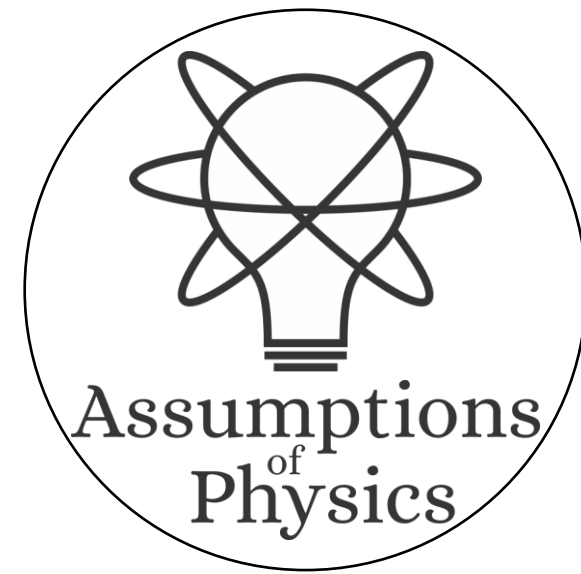
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Physics Department
University of Michigan

Centennial Workshop on Quantum Probability,
Causality, and Information

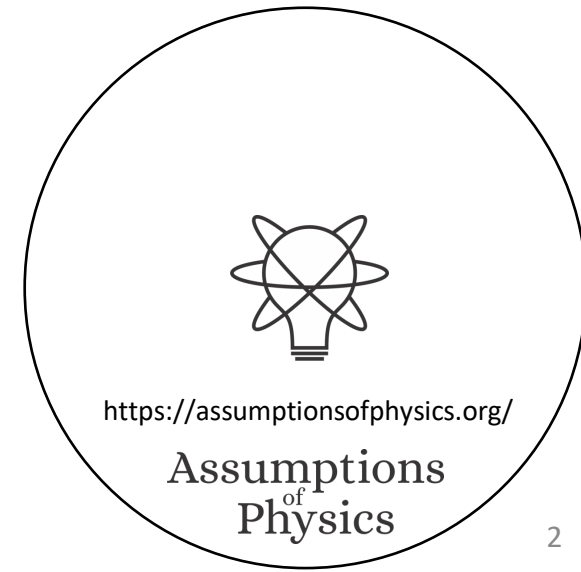
Harvard University

September 27, 2026



Outline

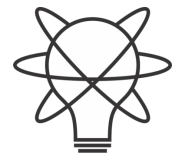
- Goal: understand exactly if/how classical and quantum probabilistic and informational structures differ, and what is a general structure that fits all physical theories
- First: deconstruct quantum mechanics to have a clear understanding of what the mathematical structures represent
- Second: start from physical requirements to develop a basic structure that must be present in all physical theories



Bunch of things bundled together

Condition QSTO:H (Quantum states and observables). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

How do we analyze this properly?



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Assumptions
of
Physics

Break physics into different conditions and study their logical relationships

Similar to what is done in the foundations of mathematics



“When the math is derived from the right physical assumptions, the physical assumptions can be derived from the math.”

Theorem 1.5. Over [CST-1D](#), the following are equivalent:

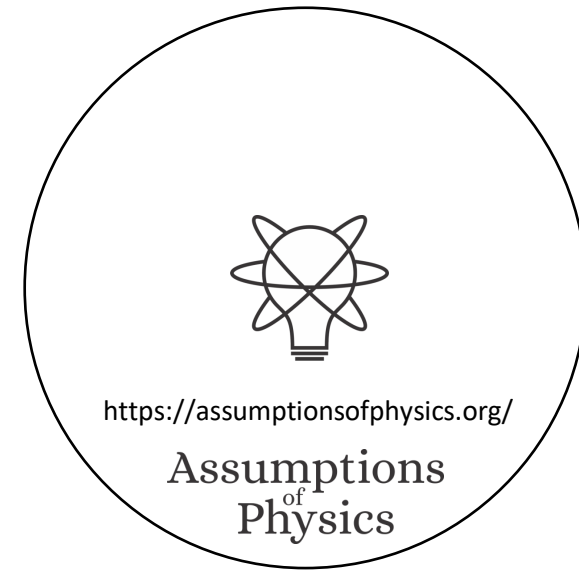
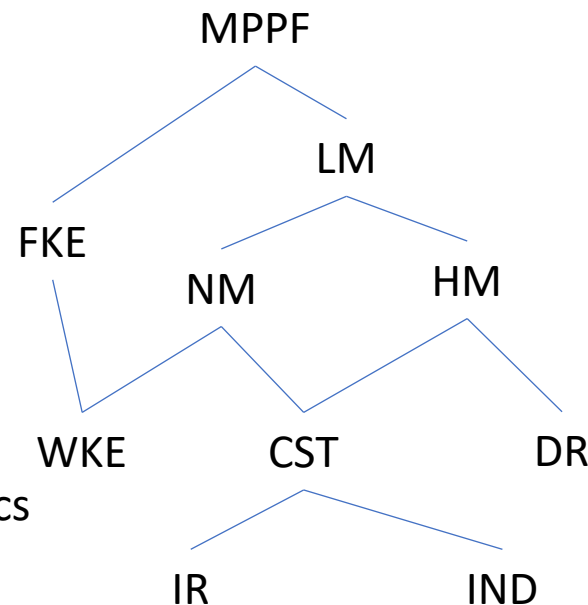
- HM-1D** (Hamiltonian mechanics) The equations of motion are given by $d_t q = \partial_p H$ and $d_t p = -\partial_q H$.
- DI-SYMP** (Symplectomorphism) The evolution leaves the symplectic form $\omega = dq \wedge dp$ invariant;
- DI-POI** (Poisson bracket invariance) The evolution leaves the Poisson bracket $\{f, g\} = \partial_q f \partial_p g - \partial_p f \partial_q g$ invariant;
- DR-DIV** (Incompressible flow) The displacement field $S^a = (d_t q, d_t p)$ is divergenceless: $\partial_a S^a = 0$;
- DR-DEN** (Volume conservation) The Liouville measure is invariant;
- DR-EV** (Determinism and reversibility) Past and future states are mapped one to one (i.e. the count of states is preserved);
- DR-THER** (Thermodynamic reversibility) The evolution is thermodynamically reversible (i.e. the entropy given by $\log \mu(U)$ is conserved);
- DR-INFO** (Conservation of information) The information about the system is conserved (i.e. the information entropy given by $-\int \rho \log \rho dq dp$ is conserved).

Found. Phys. **52**, 40 (2022)
<https://arxiv.org/abs/2608.27543>

Core assumptions
for classical mechanics

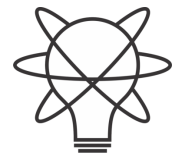
Theorem 1.4. Over Zermelo-Fraenkel set theory, the following are equivalent:

- AoC** (Axiom of Choice) For every non-empty set X there exists a function f that assigns to each nonempty set $S \subseteq X$ some representative element $f(S) \in S$;
- WOP** (Well-Ordering Principle) Every set can be well-ordered;
- ZL** (Zorn's Lemma) Every non-empty partially ordered set P with the property that every chain in P is bounded has a maximal element.



Condition QSTO:H (Quantum states and observables). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

Note: projective space does not technically require an inner product

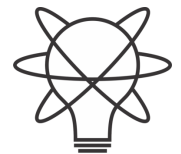


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Assumptions
of
Physics

Do projective Hilbert spaces pick QM?

1. Yes. Only quantum state spaces are represented by projective Hilbert spaces.
2. No. Some classical state spaces can also be represented in that way.
3. It's just a bunch of math. I care about physics!
4. I have no idea

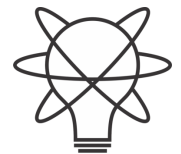


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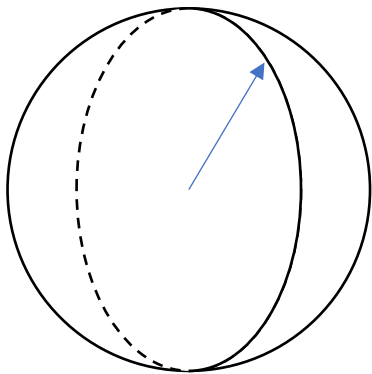
Assumptions
of
Physics

Condition QP-RAY (Quantum state space only). *The state space of a quantum system is represented by the projective space $\mathcal{P}(V)$ of a complex topological vector space V .*

Condition CP (Classical state space only). *The state space of a classical system is represented by a symplectic manifold (X, ω) .*

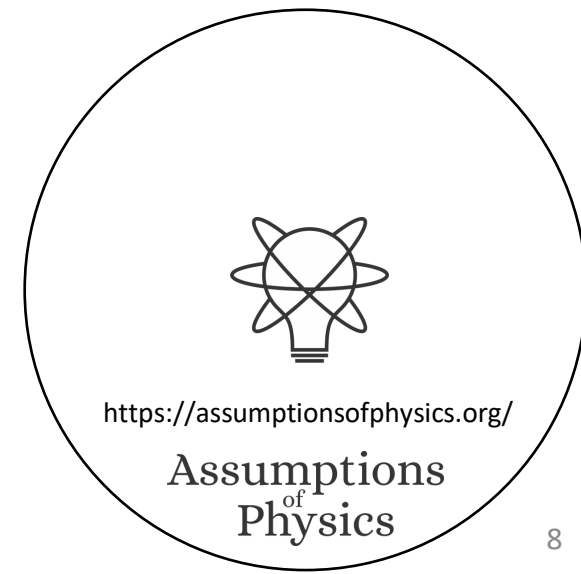
Condition CQE (Classical-Quantum exclusivity). *A system cannot be both classical and quantum.*

Theorem 5. *Condition **QP-RAY**+**CP** is consistent with $\neg$ **CQE**.*



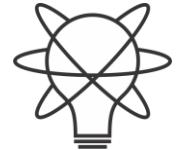
State space of a classical magnetic dipole is a sphere,
which is the complex projective line

In general, every finite-dimensional complex projective space is a symplectic manifold, and therefore allows classical Hamiltonian mechanics



Does the space of ensembles distinguish CM from QM?

1. Yes.
2. No. Some quantum systems allow representation by classical probability (e.g. Husimi Q representation).
3. Experiments distinguish between theories
4. I have no idea

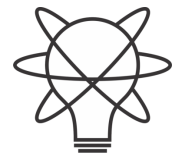


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Assumptions
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Condition QE-DENS (Quantum ensemble space). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$.*

Condition CE (Classical ensemble space). *The state space of a classical system is represented by a symplectic manifold (X, ω) . The ensemble space is represented by the set of probability distributions $\rho(q^i, p_j)$ (i.e. Radon–Nikodym derivatives of probability measures that are absolutely continuous with respect to the Liouville measure).*

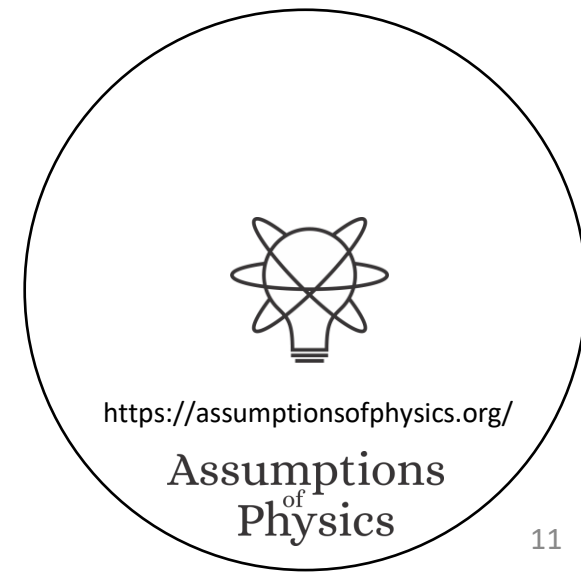
Theorem 6. *Condition **QE-DENS**+**CE** implies **CQE** for every non-trivial system (i.e. the state space has more than one state).*

In classical mechanics, every ensemble is identified by a UNIQUE probability measure on pure states (i.e. ensemble space is a simplex)

Though we lack a definition of simplex that works in the infinite-dimensional case

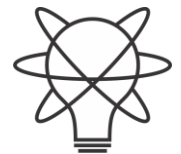
In quantum mechanics, there is NO unique probability measure

The ensemble space distinguishes
between classical and quantum mechanics



Is the ensemble space enough to characterize QM?

1. Yes. Knowing how ensembles combine during mixtures is enough.
2. No. You need an additional structure.
3. This is going nowhere. It's all about mathematical structures. Where is the physics?
4. I have no idea

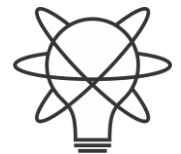


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Assumptions
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Condition [B]PMIX (Probabilities of statistical mixtures). *The conditional probability of a statistical mixture is the statistical mixture of the conditional probabilities. That is, $p(\psi \mid \sum_i \lambda_i \rho_i) = \sum_i \lambda_i p(\psi \mid \rho_i)$.*

Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in \mathcal{P}$ be a countable sequence of states. Then the following are equivalent:*

- ψ_i are mutually exclusive
- $p(\psi_i \mid \psi_j) = \delta_{ij}$
- the entropy obeys $S(\sum_i p_i \psi_i) = -\sum_i p_i \log p_i$ for all p_i such that $\sum_i p_i = 1$.

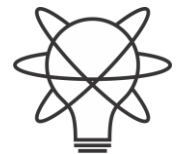
Condition BR-BORN (Conditional expectation from Born rule). *The conditional probability $p : \mathcal{P} \times \mathcal{P} \rightarrow [0, 1]$ is given by the Born rule. That is, $p(\psi \mid \phi) = \frac{\langle \phi \mid \psi \rangle \langle \psi \mid \phi \rangle}{\langle \psi \mid \psi \rangle \langle \phi \mid \phi \rangle}$.*

Condition BR-ENT (Von Neumann entropy). *The entropy function $S : \mathcal{D} \rightarrow [0, +\infty]$ is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

Condition BR-ORME (Orthogonality is mutual exclusivity). *Orthogonal states are mutually exclusive.*

Theorem 7. *Over $\text{QE-DENS} + [\text{B}] \text{MUTEX} + [\text{B}] \text{PMIX}$, conditions BR-BORN , BR-ENT and BR-ORME are equivalent. Moreover, BR-BORN is independent of $\text{QE-DENS} + [\text{B}] \text{MUTEX} + [\text{B}] \text{PMIX}$.*

The fact that orthogonal vectors
represent mutually exclusive ensembles
is an additional assumption



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Assumptions
of
Physics

QP-RAY – Pure states are rays of a complex Hilbert space

Can be classical

+

QE-DENS – Ensembles are density matrixes

Cannot be classical

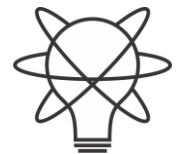
+

BR-BORN – Orthogonal states are mutually exclusive

Must be quantum

=

QST:H – Quantum states on a Hilbert space

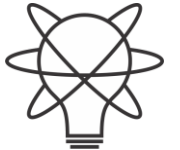


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Assumptions
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Physics

Can we reconcile evolution and measurements?

1. No. They have irreconcilable mathematical characteristics.
2. Yes. There is no problem.
3. It depends on the interpretation... and I don't care about the interpretation because I care about the physics
4. I have no idea

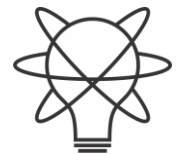


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Assumptions
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Physics

Condition DR-SCEQ (Schrödinger equation). The evolution follows the equation $\frac{d}{dt}\psi(t) = \frac{H(t)}{i\hbar}\psi(t)$, where $H(t)$ is a self-adjoint operator.

Condition DR-UNIT (Unitary evolution). Time evolution is represented by a strongly continuous one-parameter group of unitary operators. That is, $\Phi_t(\rho) = U_t\rho U_t^\dagger$, where $U_t : \mathcal{H} \rightarrow \mathcal{H}$, $U_0 = I$, $U_{t+s} = U_t U_s$ and $U_t^\dagger U_t = U_t U_t^\dagger = I$.

Condition DR-EV (Bijective map). Distinct past states map to distinct future states and vice-versa. That is, if ψ_I and ϕ_I are two initial states, ψ_F and ϕ_F are the corresponding evolved future states and ψ_P and ϕ_P are the corresponding reconstructed past states, then $\psi_I = \phi_I \iff \psi_F = \phi_F \iff \psi_P = \phi_P$.

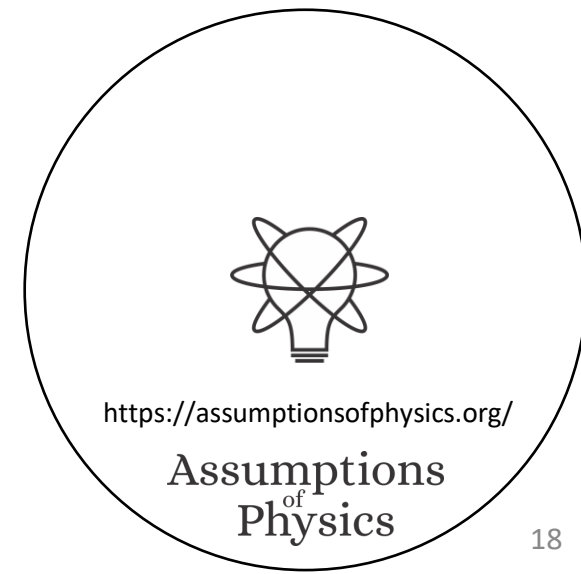
Condition DR-PROB (Preservation of conditional probability). Conditional probability remains the same across past and future states. That is, if ψ_I and ϕ_I are two initial states, ψ_F and ϕ_F are the corresponding evolved future states and ψ_P and ϕ_P are the corresponding reconstructed past states, then $p(\psi_I | \phi_I) = p(\psi_F | \phi_F) = p(\psi_P | \phi_P)$.

Condition DR-INFO (Conservation of information entropy). Future and past ensembles have the same information entropy. That is, if ρ_I is an initial ensemble, ρ_F is the corresponding evolved future ensemble and ρ_P is the corresponding reconstructed past ensemble, then $S(\rho_I) = S(\rho_F) = S(\rho_P)$.

Theorem 8. Over $QSTO:H+[B]EVMIX+EVTI$, conditions **DR-SCEQ**, **DR-UNIT**, **DR-EV**, **DR-PROB** and **DR-INFO** are equivalent.

[B]EVMIX – The evolution of the mixture is the mixture of the evolution

Unitary evolution is determinism and reversibility



Condition EQ-MEAS (Nonselective projective measurement). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x xP_x$ returns the output of a nonselective projective measurement. That is, $\Phi_X(\rho) = \sum_x P_x \rho P_x$.*

Condition EQ-LIND (Spectral equilibration). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x xP_x$ can be characterized as the equilibration under a Lindblad equation with vanishing Hamiltonian and jump operator X . That is, $\frac{d\rho}{dt} = X\rho X - \frac{1}{2}\{X^2, \rho\}$ and $\Phi_X(\rho) = \lim_{t \rightarrow \infty} \rho(t)$, where $\rho = \rho(0)$.*

Condition EQ-HS (Closest commuting ensemble). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x xP_x$ returns the ensemble commuting with X that is closest to the incoming ensemble in Hilbert–Schmidt distance. That is, $\Phi_X(\rho) = \operatorname{argmin}_{\sigma \in \mathcal{D}(\mathcal{H}), [\sigma, X]=0} \|\rho - \sigma\|_{HS}^2$.*

Condition EQ-KL (Closest commuting ensemble in information). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x xP_x$ returns the ensemble commuting with X that is closest to the incoming ensemble in relative entropy. That is, $\Phi_X(\rho) = \operatorname{argmin}_{\sigma \in \mathcal{D}(\mathcal{H}), [\sigma, X]=0} D(\rho \parallel \sigma)$.*

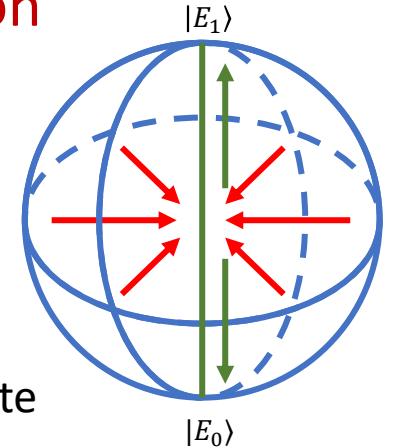
Condition [B]FINENT (Entropy is finite). *All ensembles have a well-defined finite entropy. Therefore, mathematical objects representing ensembles with infinite entropy are unphysical, and should be discarded.*

Theorem 9. *Over $QSTO:H$, conditions EQ-MEAS, EQ-LIND and EQ-HS are equivalent. Over $QSTO:H+[B]FINENT$, condition EQ-KL is also equivalent to the others.*

Nonselective projective measurements are equilibrations

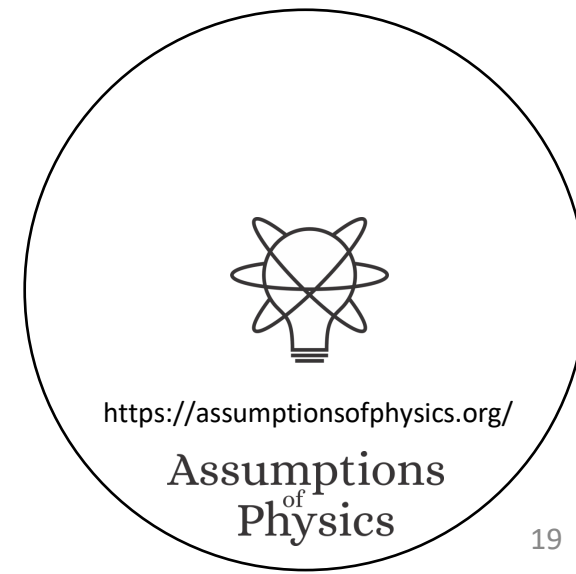
Projective measurement

Step 1: interaction



Step 2: update

same as classical update



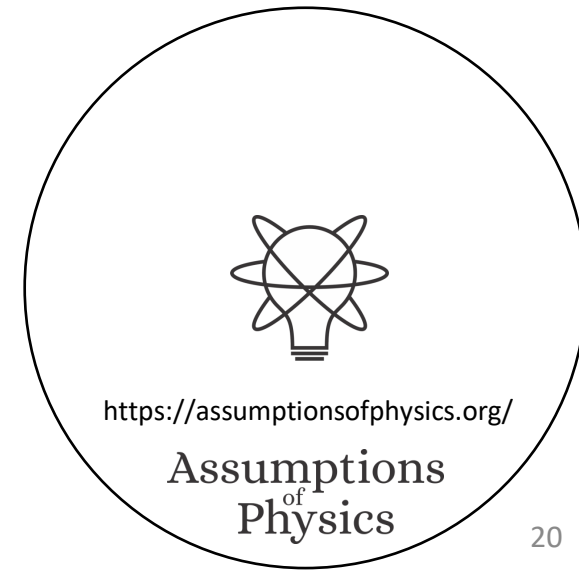
Definition 2 (State-adaptive projective process). A state-adaptive projective process is a process that acts as a projective measurement on each ensemble. That is, it is an affine map $\Psi : \mathcal{D} \rightarrow \mathcal{D}$ on the ensemble space such that, for every incoming ensemble ρ , there is a projective decomposition $\{P_a^\rho\}$ of the identity for which $\Psi(\rho) = \sum_a P_a^\rho \rho P_a^\rho$. The projective decomposition may depend on the incoming ensemble.

Condition DR-PROJ (Projective evolution). Time evolution is the limit of repeated applications of an infinitesimal state-adaptive projective process that allows perfect prediction and perfect reconstruction. That is, evolution over elapsed time t is obtained by applying the same process of duration t/n , n times, in the limit $n \rightarrow \infty$. Moreover, the probabilities of the predicted and reconstructed state sequences tend to one.

Theorem 10. Over [QSTO:H](#), conditions [DR-UNIT](#) and [DR-PROJ](#) are equivalent.

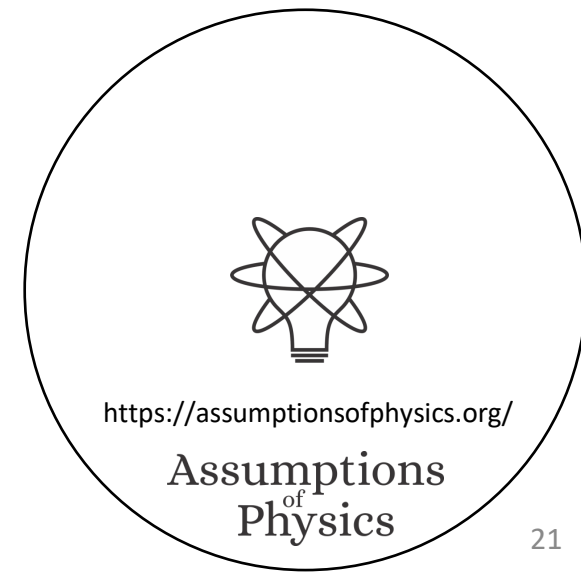
Unitary evolution is a continuous sequence of infinitesimal adaptive measurements

$$\begin{array}{ccc}
 X|\psi\rangle = x|\psi\rangle & \begin{array}{c} \uparrow \quad \nearrow \quad \nearrow \quad \nearrow \quad \nearrow \quad \nearrow \quad \nearrow \quad \nearrow \end{array} & X_t|\psi(t)\rangle = x|\psi(t)\rangle \\
 |\psi(t)\rangle = U_t|\psi\rangle & & X_t = U_t X U_t^\dagger
 \end{array}$$



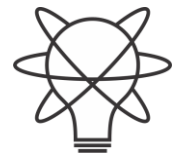
What do states represent?

1. An ontological description that exists regardless of what happens
2. An epistemic description of what the agent knows
3. It's physics! It's about stuff that I can do in the lab! You see, I put this here... and then something happens. It's about stuff that happens!
4. I have no idea



What do states represent?

1. An ontological description that exists regardless of what happens
2. An epistemic description of what the agent knows
3. **The physics! It's about stuff that I can do in the lab! You see, I put this here... and then something happens. It's about stuff that happens!**
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Assumptions
of
Physics

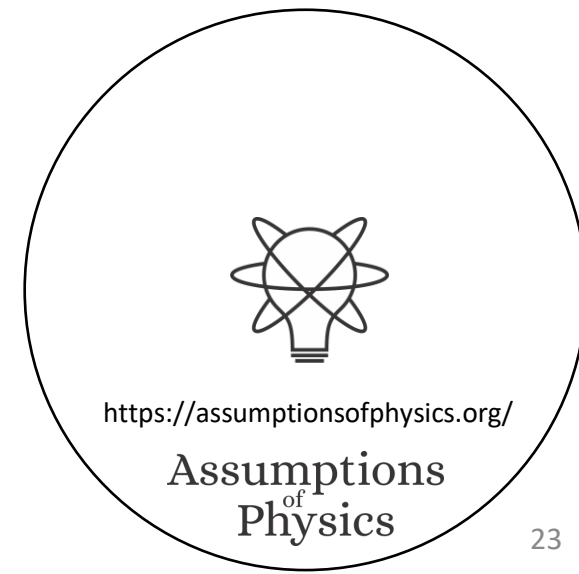
Condition EE-DYN (Dynamic equilibrium). *Every ensemble ρ is the dynamic equilibrium of some unitary process.*

Condition EE-MEAS (Spectral equilibrium). *Every ensemble ρ is the output of some spectral equilibration process (i.e. nonselective projective measurement).*

Condition EE-THER (Thermodynamic equilibrium). *Every full-rank ensemble ρ is the thermodynamic equilibrium for some Hamiltonian. Every non-full-rank ensemble is an approximate thermodynamic equilibrium for some Hamiltonian.*

Theorem 12. *Condition **QSTO:H** implies **EE-DYN**, **EE-MEAS** and **EE-THER**.*

States are the equilibrium ensembles
of some process



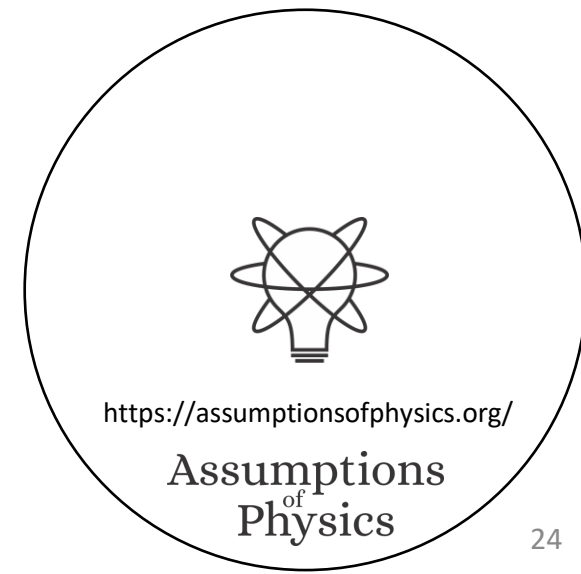
How do we turn these insights into a general theory of physics?

Every formal theory starts with primitive objects

Objects that are NOT defined from other objects

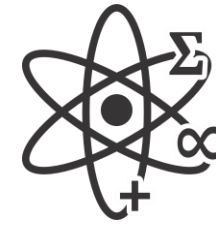
We don't want to kick the can down the road

Are ensembles the correct primitive objects for physical theories? How do we know?



Start from mathematical axioms with a clear physical justification

Axioms must represent constitutive assumptions



Physical Mathematics

*“When physical objects are mapped to the right mathematical objects,
the physical specification maps to the mathematical definitions.”*

Axiom 1.7 (Axiom of mixture). *The statistical mixture of two ensembles is an ensemble. Formally, an ensemble space $\mathcal{E}$ is equipped with an operation $+$: $[0, 1] \times \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{E}$ called **mixing**, noted with the infix notation $pa + \bar{p}b$, with the following properties:*

- **Continuity:** *the map $+(p, a, b) \rightarrow pa + \bar{p}b$ is continuous (with respect to the product topology of $[0, 1] \times \mathcal{E} \times \mathcal{E}$)*
- **Identity:** $1a + 0b = a$
- **Idempotence:** $pa + \bar{p}a = a$ for all $p \in [0, 1]$
- **Commutativity:** $pa + \bar{p}b = \bar{p}b + pa$ for all $p \in [0, 1]$
- **Associativity:** $p_1e_1 + \bar{p}_1 \left(\left(\frac{p_3}{\bar{p}_1} \right) e_2 + \frac{p_3}{\bar{p}_1} e_3 \right) = \bar{p}_3 \left(\frac{p_1}{\bar{p}_3} e_1 + \left(\frac{p_1}{\bar{p}_3} \right) e_2 \right) + p_3e_3$ where $p_1 + p_3 \leq 1$

Justification. This axiom captures the ability to create a mixture merely by selecting between the output of different processes. Let e_1 and e_2 be two ensembles that represent the output of two different processes P_1 and P_2 . Let a selector S_p be a process that outputs two symbols, the first with probability p and the second with probability $\bar{p}$. Then we can create another process P that, depending on the selector, outputs either the output of P_1 or P_2 . All possible preparations of such a procedure will form an ensemble. Therefore we are justified in equipping an ensemble space with a mixing operation that takes a real number from zero to one, and two ensembles.

Given that mixing represents an experimental relationship, and all experimental rela-

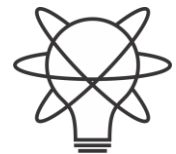
Informal intuitive requirement

(something that makes sense to a physicist or an engineer)

Formal definition

(something a mathematician will find precise)

Show that the formal definition
follows from the intuitive requirement



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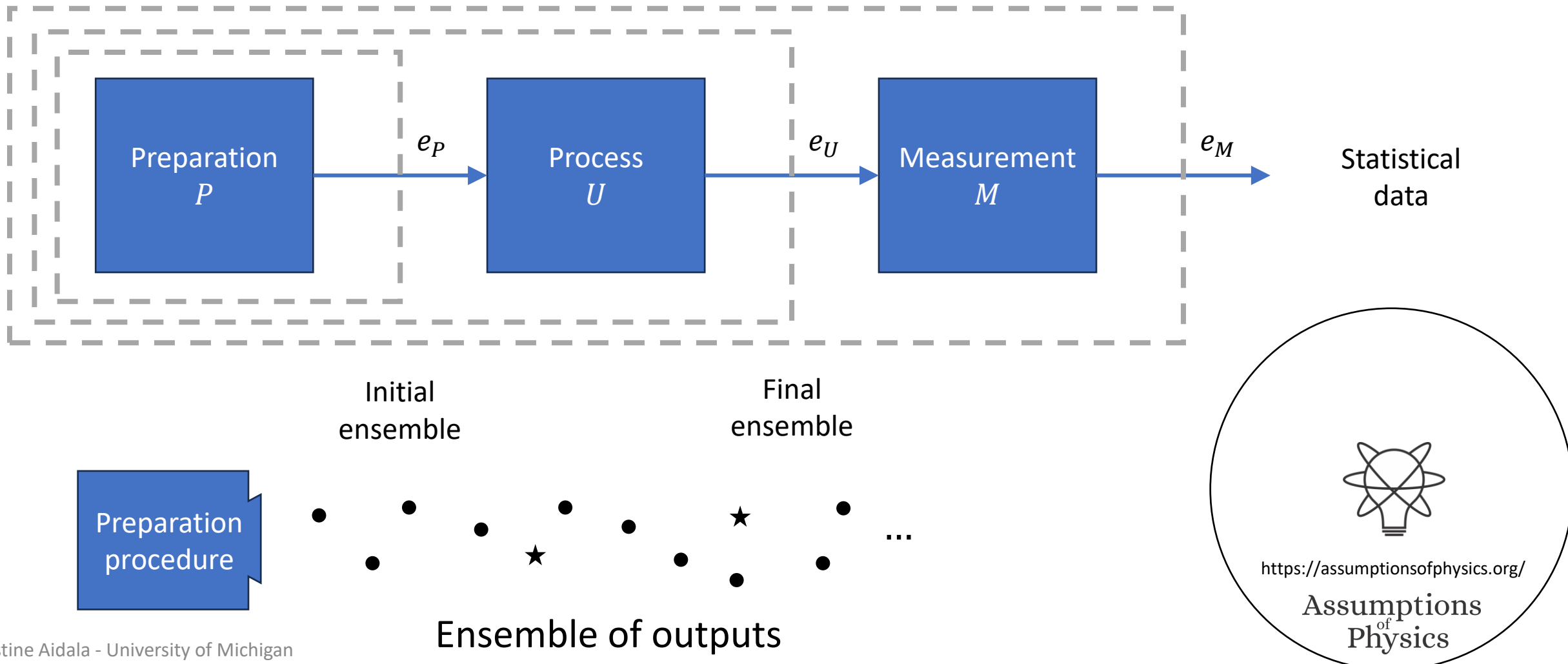
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Assumptions of Physics, Michigan Publishing (v3 2025)

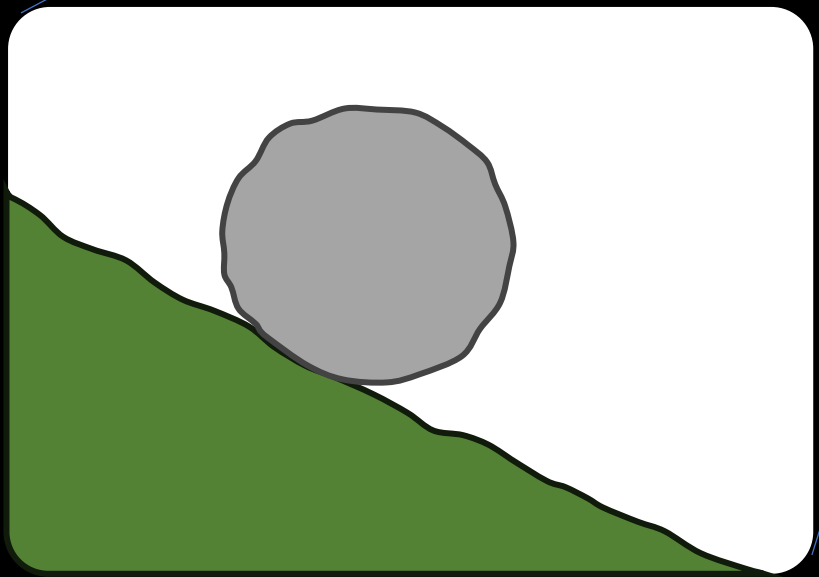
<https://doi.org/10.3998/mpub.12204707>

Principle of scientific reproducibility. Scientific laws describe relationships that can always be experimentally reproduced.

⇒ Scientific laws are relationships between ensembles

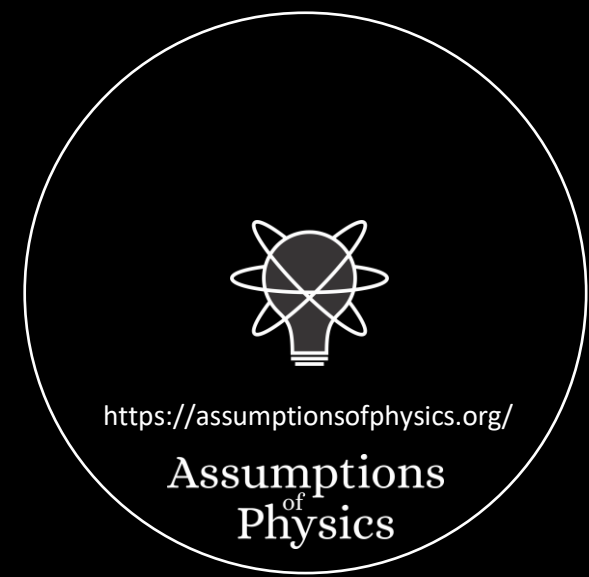


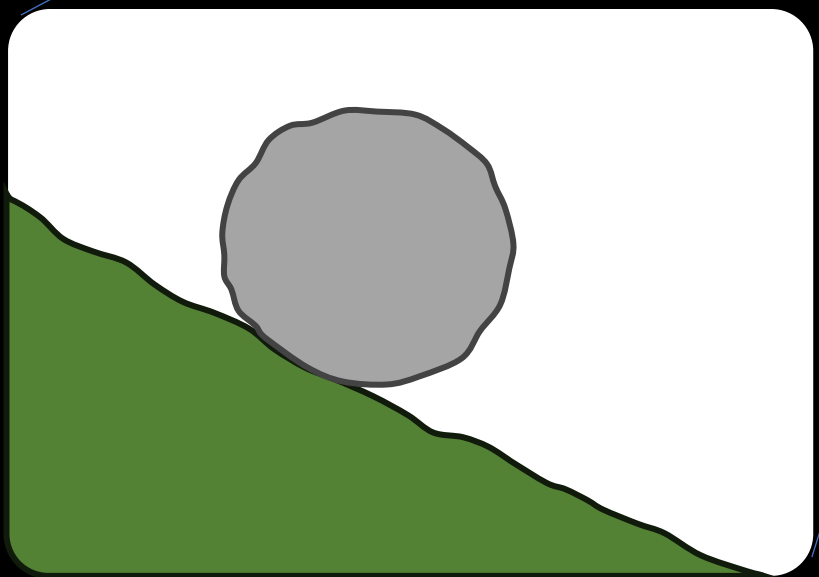
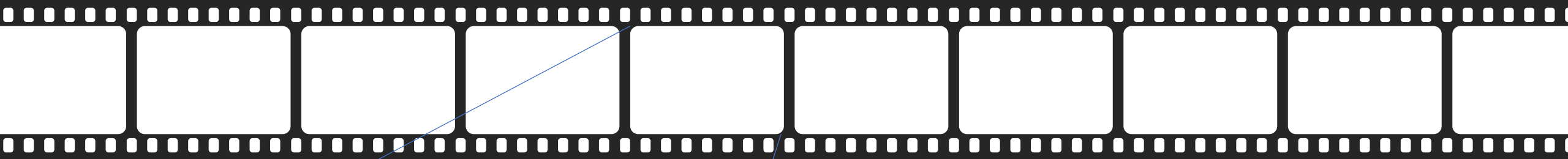
→ t



Physics describes the evolution
of a particular system like a movie

$$x(t)$$

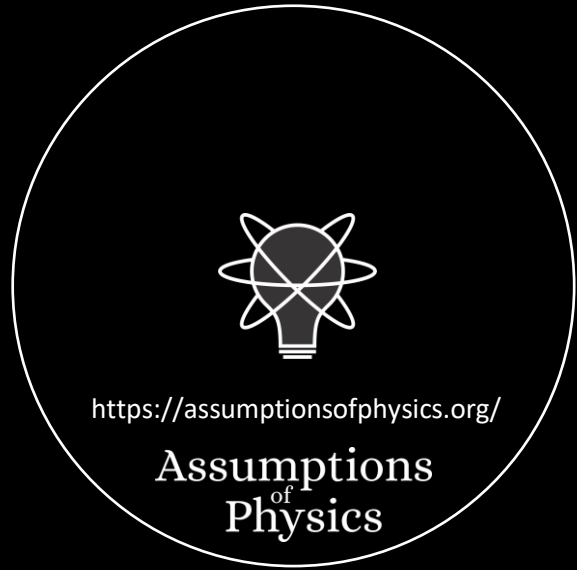




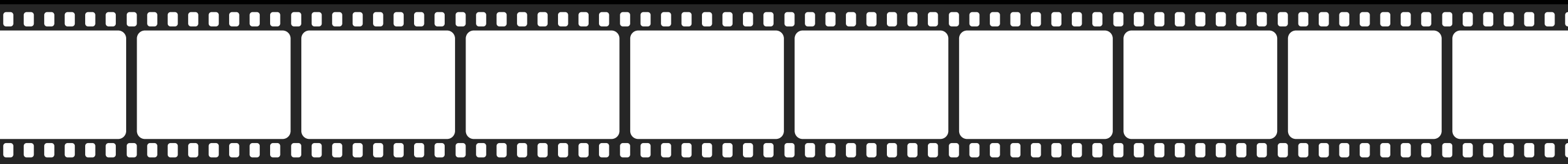
That's not it!!!

~~Physics describes the evolution
of a particular system like a movie~~

$$x(t)$$



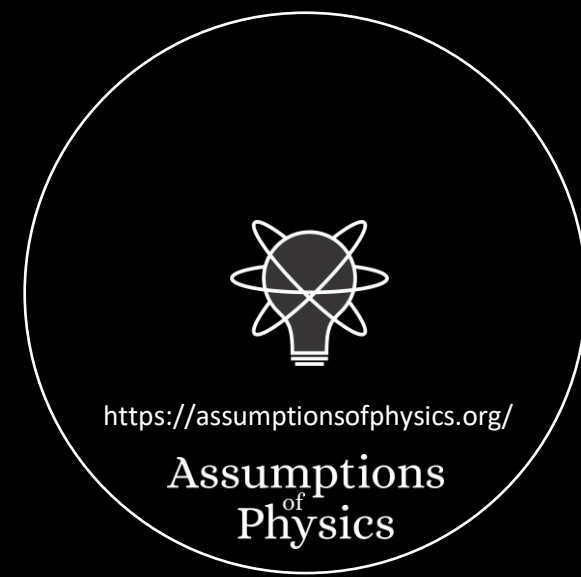
→ t



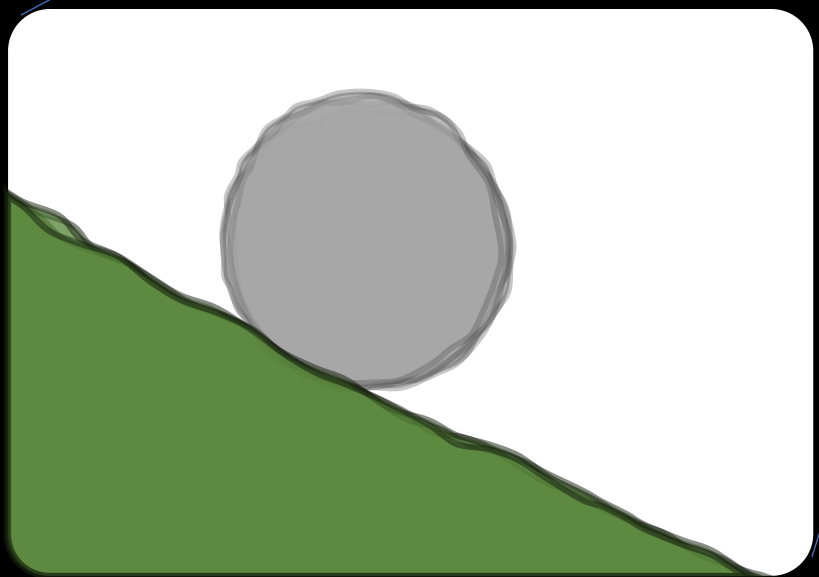
Physics describes the evolution
of similarly prepared systems

Every time I do this...

$$x(t)$$



→ t



Physics describes the evolution
of similarly prepared systems

$$x(t)$$



<https://assumptionsofphysics.org/>

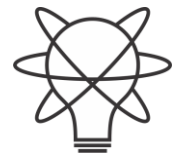
Assumptions
of
Physics

Statistical description of preparations
and measurement outcomes

$x(t)$

In physics, we can only
study relationships that
can be reproduced:
relationships between
ensembles

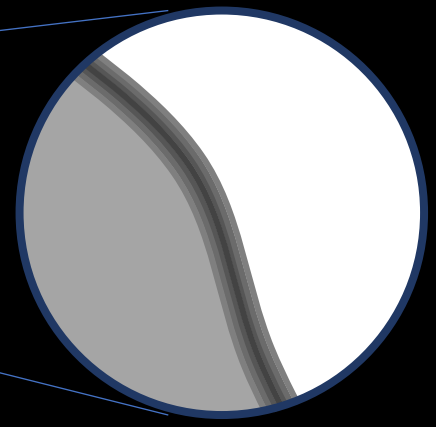
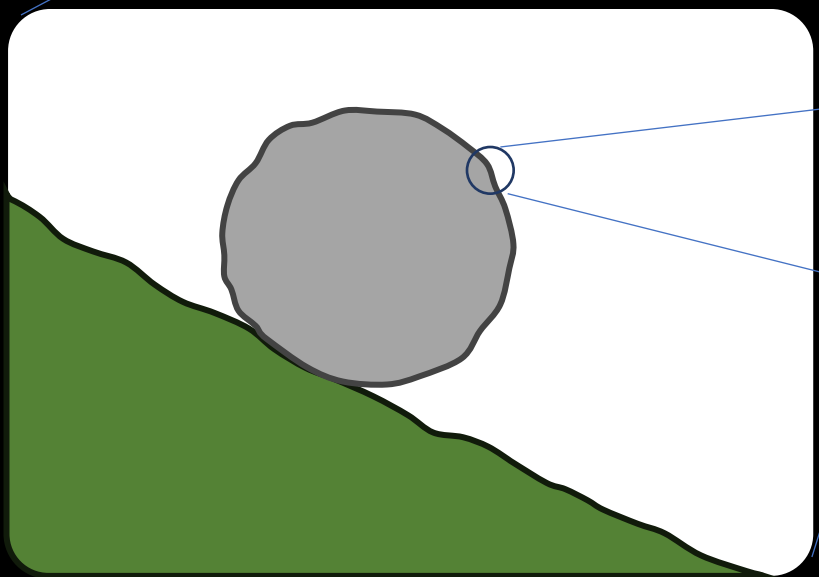
Statistical description
of all clocks synchronized
at a particular initial event



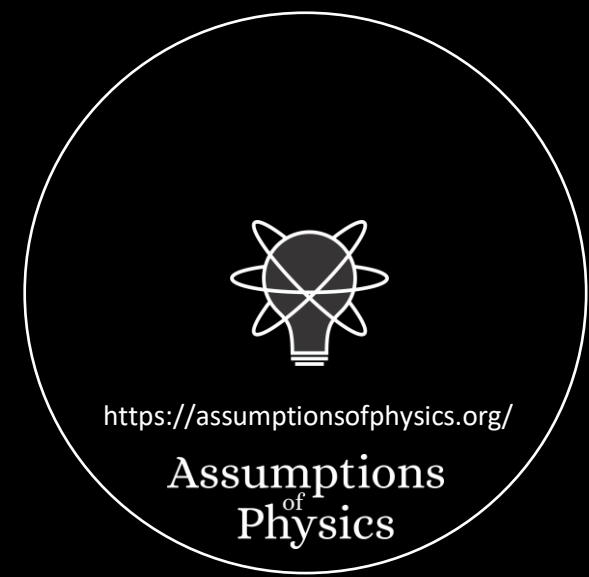
<https://assumptionsofphysics.org/>

Assumptions
of
Physics

→ t



Even a single preparation/measurement
is an average over a finite time



To be able to define a probability distribution over heads or tails, you need first to define what heads and tails are

We can do that because we can leave the coin be, and it remains heads or tails

If it kept changing more than 100 times a second, we wouldn't perceive heads or tails: just a blur

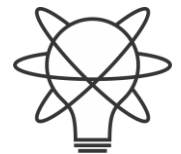


wikipedia

wikipedia

The state of a system is what remains
“stable” for at least a short amount of time

It's ensembles all the way down



<https://assumptionsofphysics.org/>

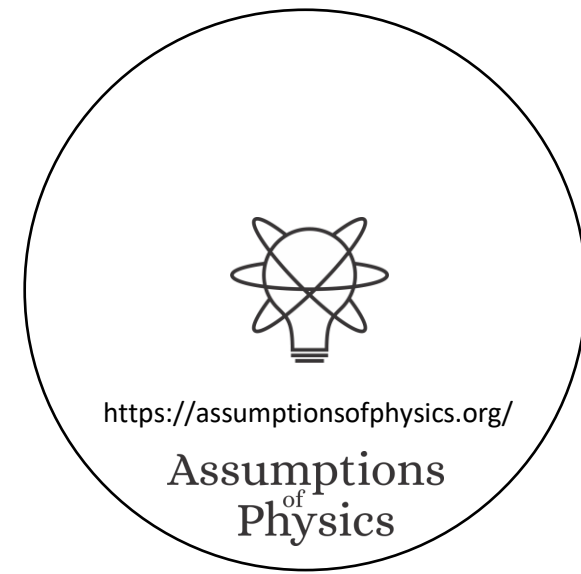
Assumptions
of
Physics

Principle of scientific reproducibility. Scientific laws describe relationships that can always be experimentally reproduced.

⇒ Scientific laws are relationships between ensembles

Ensembles MUST be the foundational primitives

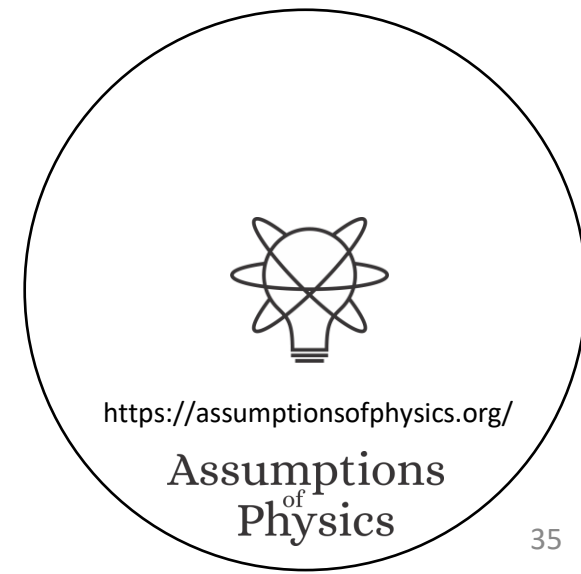
Without a generic theory of ensemble spaces there is no well-defined theory of physics, causality, probability, information theory, ...



Four foundational pillars

- Experimental verifiability: topology and σ -algebras
- Statistical mixing: convex and linear structures
- Ensemble variability: entropy and all geometric structures
- Equilibria: Lie algebraic structures and differentiability

Understanding the requirements of these structures and their relationships is the ultimate problem

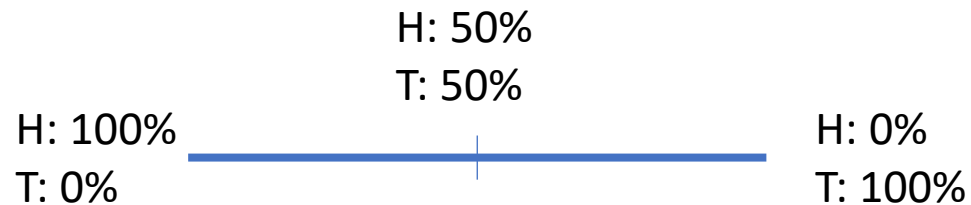


~~Mass of the photon is 0 eV~~ Not verifiable!

Verifiable!

Mass of the photon is less than 10^{-13} eV

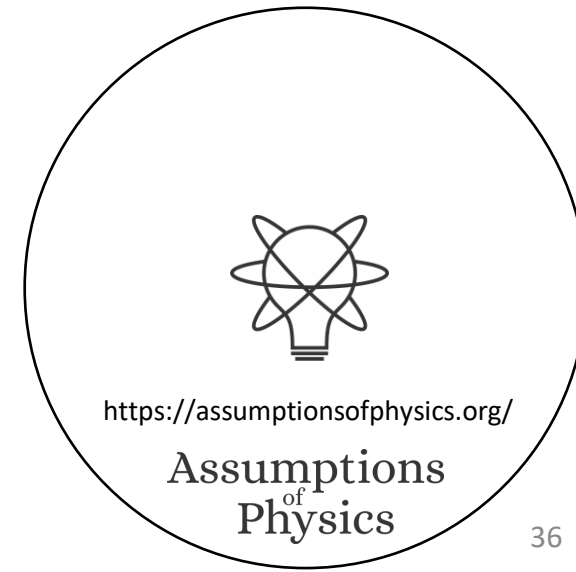
Space of coin tossers



Open set



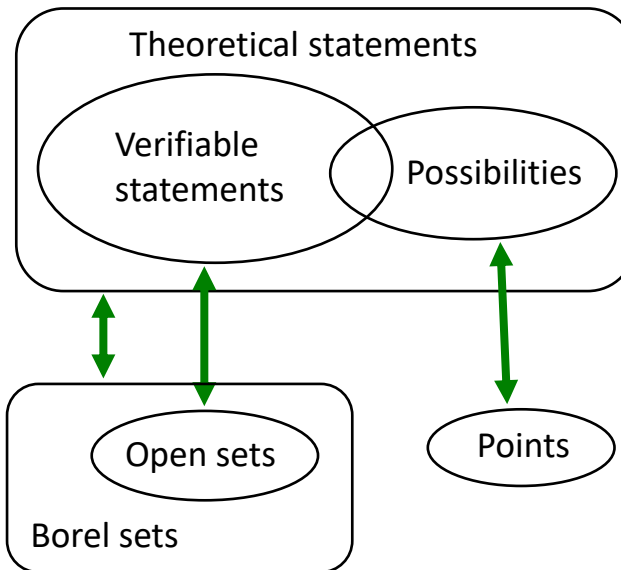
The coin is fair $\Leftrightarrow p(H) \in (0.5 - \epsilon, 0.5 + \epsilon)$



Experimental verifiability

Science is evidence based $\Rightarrow$ Statements are associated to experimental tests

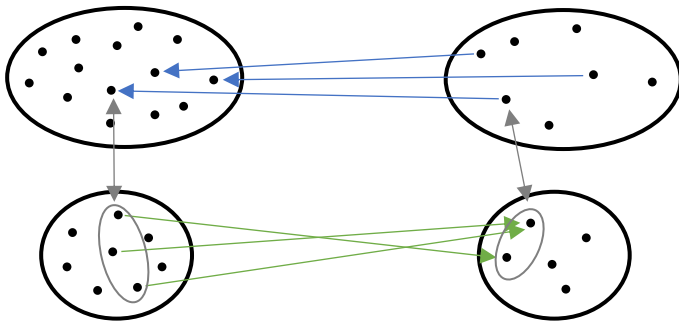
Verifiable statement	Test Result
T	SUCCESS (in finite time)
F	UNDEFINED FAILURE (in finite time)



s	Test
T	S
	U
F	U
	F

$int(A)$ corresponds to the verifiable part of a statement
 ∂A corresponds to the undecidable part of a statement
 $ext(A)$ corresponds to the falsifiable part of a statement

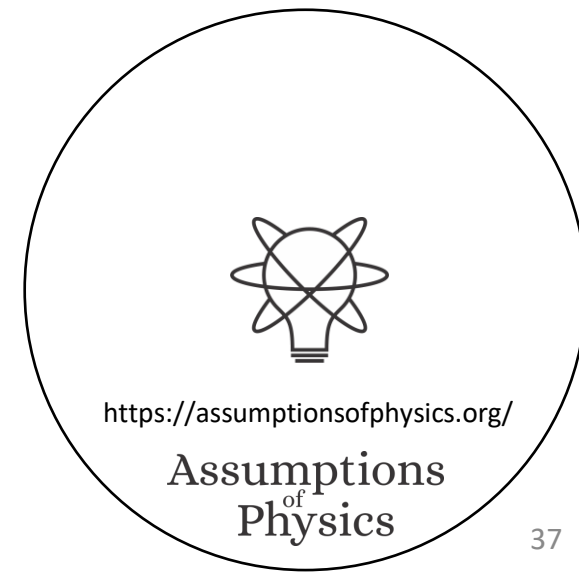
Inference relationship $r: \mathcal{D}_Y \rightarrow \mathcal{D}_X$ such that $r(s) \equiv s$



Causal relationship $f: X \rightarrow Y$ such that $x \preceq f(x)$

Experimental verifiability
 $\Rightarrow$ **topology and σ -algebra**

Experimental relationships must be
 topologically continuous



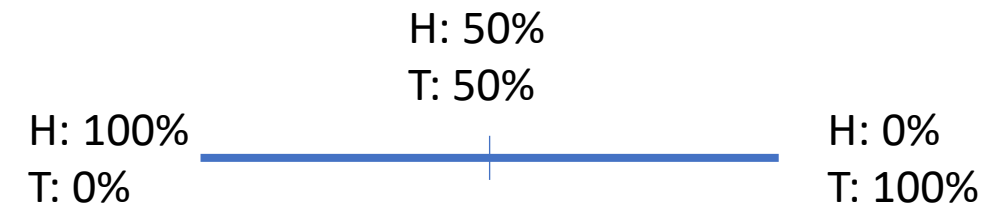
The problem with Bayesianism

This does not work!

In formal models of belief change, the belief set represents the set of full beliefs. Uncertain beliefs are not represented in these models. The most common way to represent uncertainties is to use a probabilistic model. In such models, the beliefs of an epistemic agent are represented by a probability function that assigns a real number between 0 and 1 to all propositions. This number represents the degree of belief that the epistemic agent has in the proposition.

“Logic of Belief Revision,” §10.1, Stanford Encyclopedia of Philosophy

The coin is PERFECTLY fair $p(H) = 0.5$



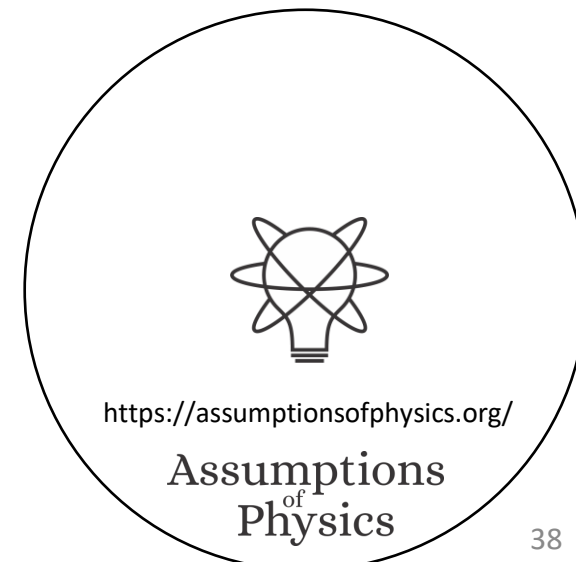
I have no idea about the fairness of the coin

~~$p(H) = 0.5$~~

$p(H) \in [0,1]$

Imprecise probability

Merging topology with imprecise probability is likely the only way to make probabilistic inference work in the continuum



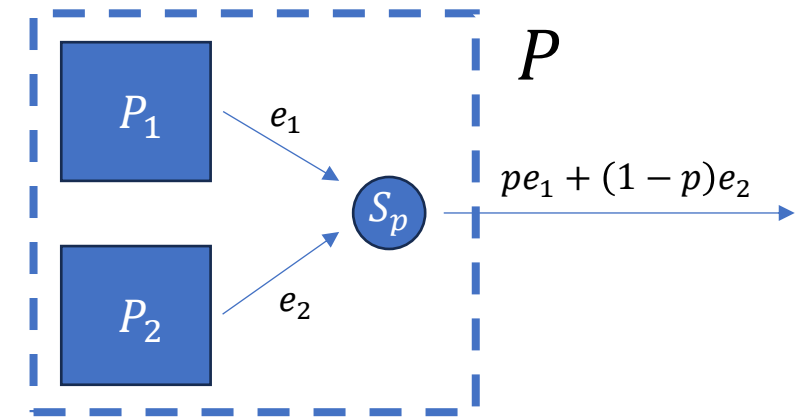
Axiom 1.4 (Axiom of ensemble). The state of a system is represented by an *ensemble*, which represents all possible preparations of equivalent systems prepared according to the same procedure. The set of all possible ensembles for a particular system is an *ensemble space*. Formally, an ensemble space is a T_0 second countable topological space where each element is called an ensemble.

Axiom 1.7 (Axiom of mixture). The statistical mixture of two ensembles is an ensemble. Formally, an ensemble space $\mathcal{E}$ is equipped with an operation $+: [0, 1] \times \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{E}$ called *mixing*, noted with the infix notation $pa + \bar{p}b$, with the following properties:

- **Continuity:** the map $+(p, a, b) \rightarrow pa + \bar{p}b$ is continuous (with respect to the product topology of $[0, 1] \times \mathcal{E} \times \mathcal{E}$)
- **Identity:** $1a + 0b = a$
- **Idempotence:** $pa + \bar{p}a = a$ for all $p \in [0, 1]$
- **Commutativity:** $pa + \bar{p}b = \bar{p}b + pa$ for all $p \in [0, 1]$
- **Associativity:** $p_1e_1 + \bar{p}_1 \left(\left(\frac{p_3}{\bar{p}_1} \right) e_2 + \frac{p_3}{\bar{p}_1} e_3 \right) = \bar{p}_3 \left(\frac{p_1}{\bar{p}_3} e_1 + \left(\frac{p_1}{\bar{p}_3} \right) e_2 \right) + p_3e_3$ where $p_1 + p_3 \leq 1$

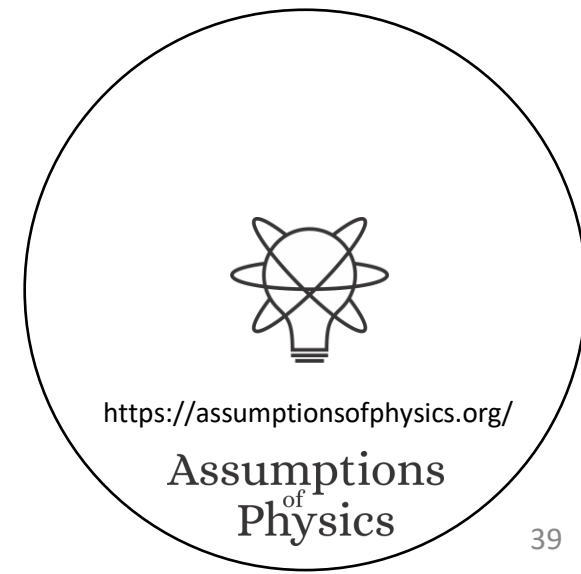
Justification. This axiom captures the ability to create a mixture merely by selecting between the output of different processes. Let e_1 and e_2 be two ensembles that represent the output of two different processes P_1 and P_2 . Let a selector S_p be a process that outputs

Ensembles can be mixed, and the mixing operation must preserve experimental verifiability (i.e. topologically continuous)



$\Rightarrow$ Topological convex space

There is no established theory!

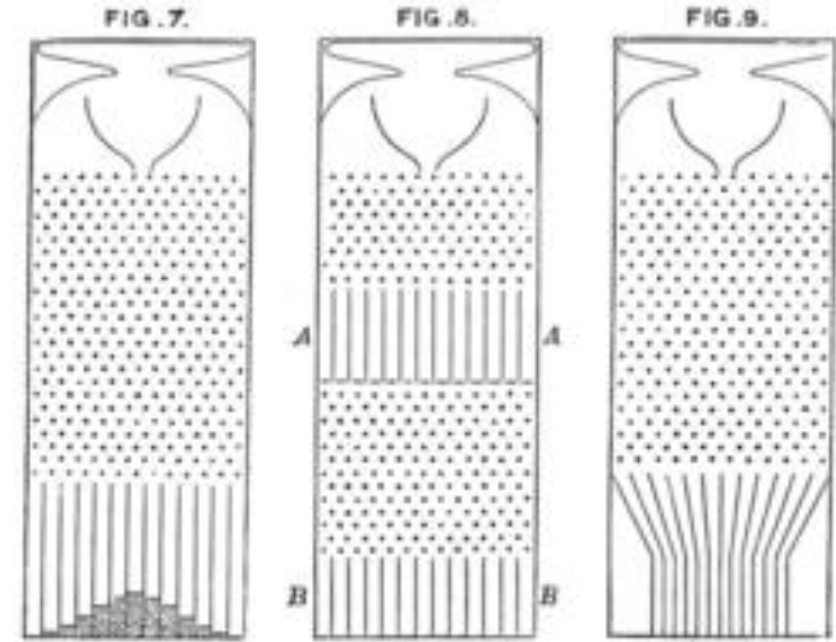


Note: mixing coefficients are NOT probabilities

In general, they are parameters of the preparation procedure

They apply recursively to other ensembles, not to events

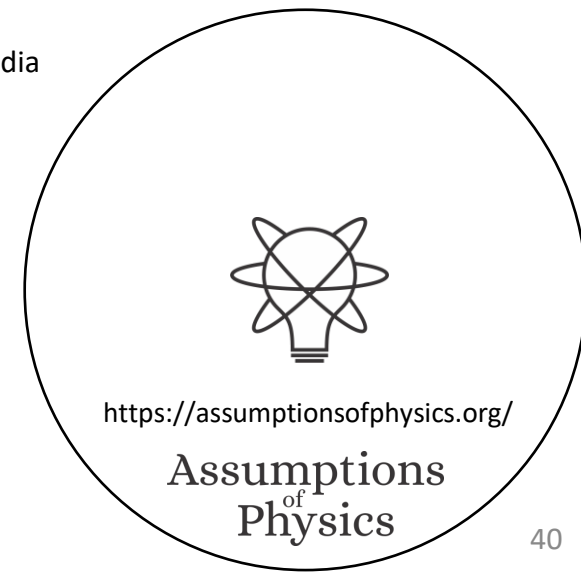
$$p_1(p_2a + \overline{p_2}b) + \overline{p_1}(p_3a + \overline{p_3}c)$$



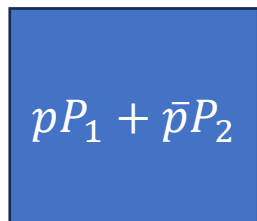
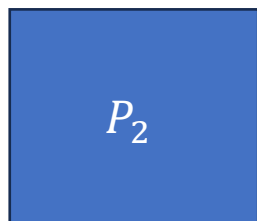
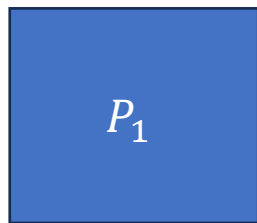
Galton's original drawings - wikipedia

Different interpretation of probability

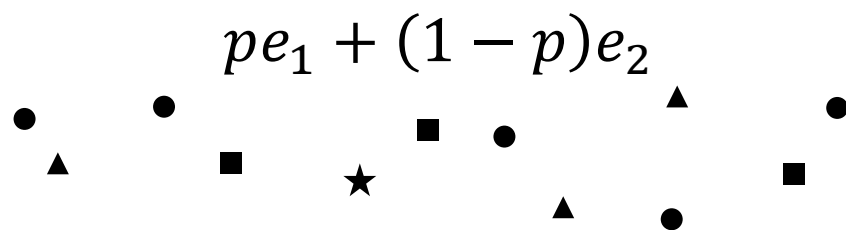
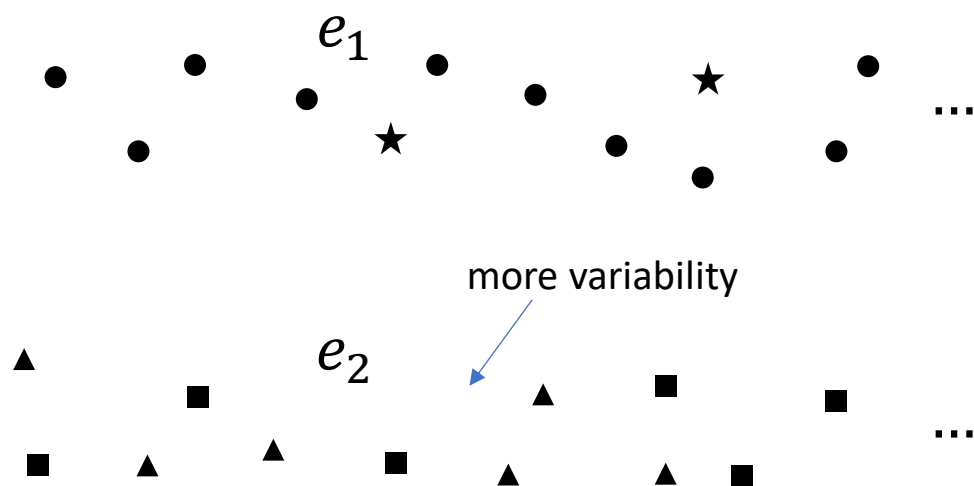
Recovers standard probability in special cases,
likely with propensities for ensembles and frequencies for elements



Preparation



Ensemble



Entropy



Variability within an ensemble

$$S(e_1)$$

One instance is enough to tell e_1 and e_2 apart

Mutually exclusive preparations

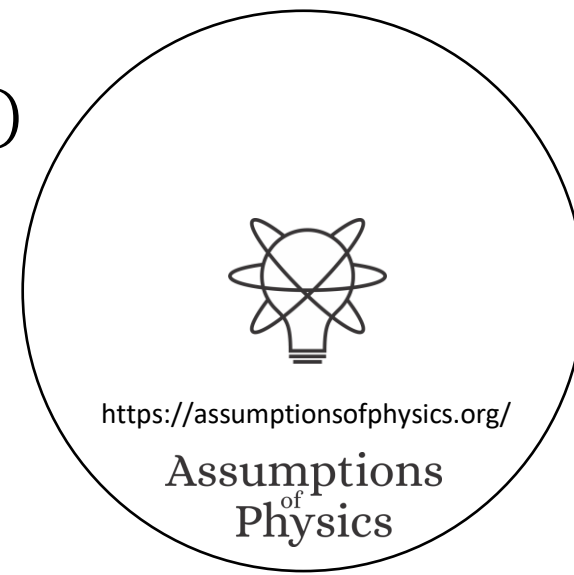
$$S(e_2)$$

Orthogonal



Maximal entropy increase if orthogonal

$$\leq pS(e_1) + \bar{p}S(e_2) + I(p, \bar{p})$$



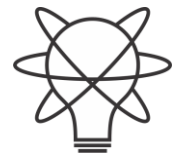
Axiom 1.58 (Axiom of entropy). *Every ensemble is associated with an **entropy** which quantifies the variability of the instances within an ensemble. Formally, an ensemble space $\mathcal{E}$ is equipped with a function $S : \mathcal{E} \rightarrow \mathbb{R}$, defined up to a positive multiplicative constant representing the unit numerical value. The entropy has the following properties:*

- **Continuity**^a
- **Strict concavity**: $S(pa + \bar{p}b) \geq pS(a) + \bar{p}S(b)$ with the equality holding if and only if $a = b$ or $p \in \{0, 1\}$
- **Upper variability bound**: there exists a universal function $I(p, \bar{p})$ (i.e. the same for all ensemble spaces) such that $S(pa + \bar{p}b) \leq I(p, \bar{p}) + pS(a) + \bar{p}S(b)$; if the equality holds for all $p \in [0, 1]$, a and b are **mutually exclusive** or **orthogonal**, noted $a \perp b$
- **Mixtures preserve orthogonality**:^b $a \perp b$ and $a \perp c$ if and only if $a \perp pb + \bar{p}c$ for any $p \in (0, 1)$

Define entropy axiomatically, just based on bounds and properties

Full justification given

Axiomatization likely does not capture all requirements



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Assumptions
of
Physics

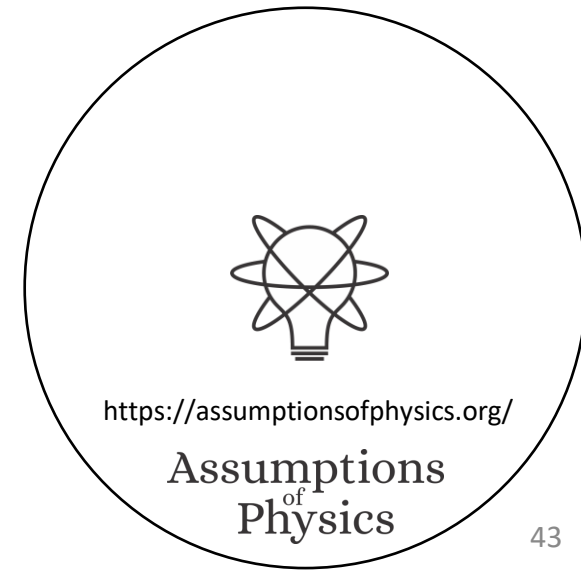
The entropy upper bound $I(p, \bar{p})$ is uniquely determined

Theorem 1.25 (Uniqueness of entropy). *The entropy of the coefficients $I(p, \bar{p})$ is the Shannon entropy. That is, $I(p, \bar{p}) = -\kappa (p \log p + \bar{p} \log \bar{p})$ where $\kappa > 0$ is the arbitrary multiplicative constant for the entropy. For a mixture of arbitrarily many elements, $I(\{p_i\}) = -\kappa \sum_i p_i \log p_i$.*

Shannon entropy

Maximal increase in variability during mixing

Proof “does not know” whether we are dealing with classical ensembles, quantum ensembles, or ensembles for a theory yet to be discovered



Two ways to define “exclusive” ensembles

Separate ensembles

$$a \perp b$$

No “common component” $c \in \mathcal{E}$

such that

$$a = p_1 c + \bar{p}_1 e_1$$

$$b = p_2 c + \bar{p}_2 e_2$$



Orthogonal ensembles

$$a \perp b$$

Saturate upper entropy bound

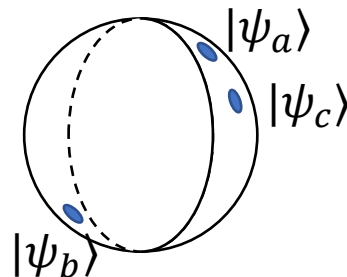
$$S(pa + \bar{p}b) = I(p, \bar{p}) + pS(a) + \bar{p}S(b)$$

Coincide in classical
ensemble spaces

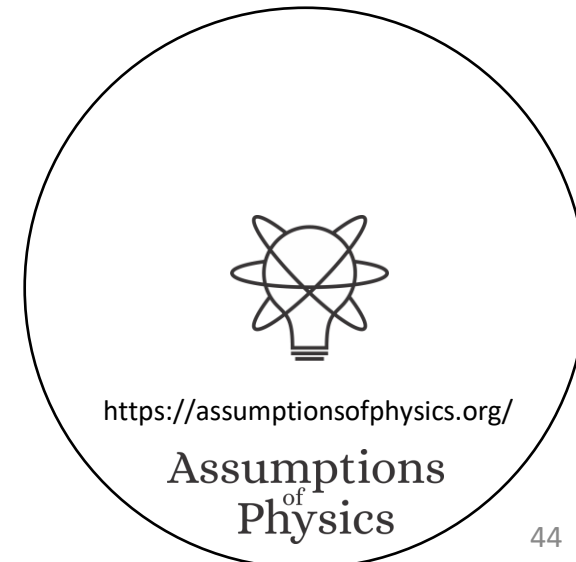


disjoint support

Different in quantum
ensemble spaces



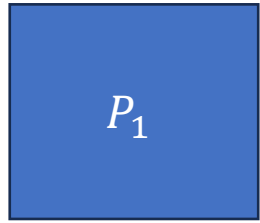
Simple but powerful
definitions from
basic axioms



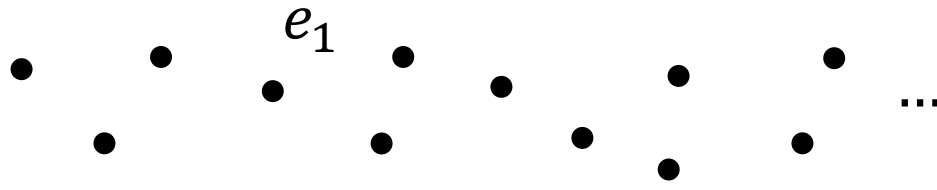
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Assumptions
of
Physics

Preparation

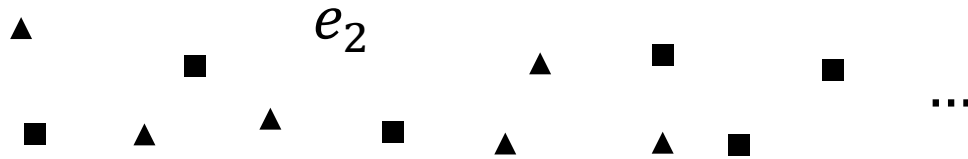
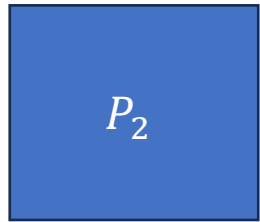


Ensemble



Identically
prepared
ensemble

There is no variability
in the ensemble



Pure
ensemble

No preparation with
lower variability

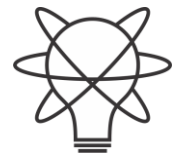
There is no a such that $e_2 = pa + \bar{p}b$ for some $p \in (0,1)$ and $b \neq a$

In classical theories, all pure ensembles are identically prepared

No real distinction between pure preparations and outcomes

In non-classical theories, no identically prepared ensembles

Zero entropy does NOT correspond to identically prepared ensembles



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Assumptions
of
Physics

What is classical probability?

All elements of the sample space must be mutually exclusive

Classical probability can only be defined among mutually exclusive **preparations** (i.e. orthogonal ensembles)

$$e = \sum p_i e_i \quad e_i \perp e_j \quad \forall i \neq j$$

probability $\Leftrightarrow$ decomposition into orthogonal ensembles

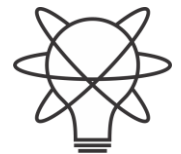
Despite the fact that Q is non-negative definite and bounded like a standard [joint probability distribution](#), this similarity may be misleading, because different coherent states are not orthogonal. Two different points α do not represent disjoint physical contingencies; thus, $Q(\alpha)$ does *not represent the probability of mutually exclusive states*, as needed in the [third axiom of probability theory](#).

Husimi Q representation is NOT probability

Conditions of a sample space [\[edit\]](#)

A set Ω with outcomes $s_1, s_2, \dots, s_n$ (i.e. $\Omega = \{s_1, s_2, \dots, s_n\}$) must meet some conditions in order to be a sample space.^[13]

- The ~~outcomes~~ **preparations!** must be **mutually exclusive**, i.e. if s_j occurs, then no other s_i will take place, $\forall i, j = 1, 2, \dots, n \quad i \neq j$.^[6]
- The outcomes must be **collectively exhaustive**, i.e. on every experiment (or random trial) there will always take place some outcome $s_i \in \Omega$ for $i \in \{1, 2, \dots, n\}$.^[6]



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Assumptions
of
Physics

How much does the entropy increase during mixture?

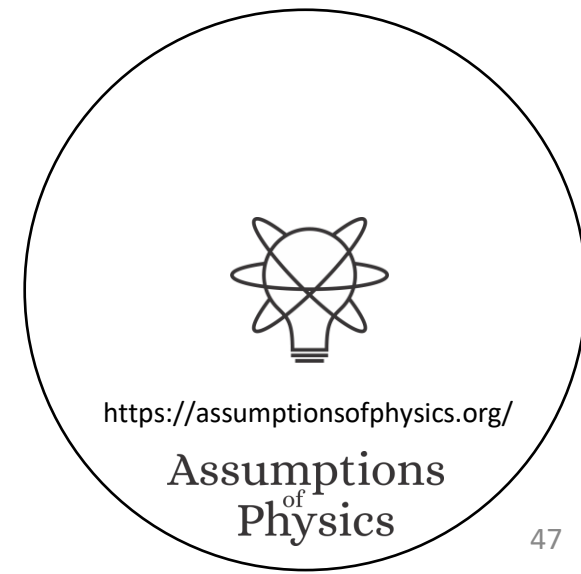
$$MS(a, b) = S\left(\frac{1}{2}a + \frac{1}{2}b\right) - \left(\frac{1}{2}S(a) + \frac{1}{2}S(b)\right)$$

1. *non-negativity*: $MS(a, b) \geq 0$
2. *identity of indiscernibles*: $MS(a, b) = 0 \iff a = b$
3. *unit boundedness*: $MS(a, b) \leq 1$
4. *maximality of orthogonals*: $MS(a, b) = 1 \iff a \perp b$
5. *symmetry*: $MS(a, b) = MS(b, a)$

Recovers the Jensen-Shannon divergence (JSD)
(both classical and quantum)

Pseudo-distance defined from the entropy

(does not satisfy the triangle inequality)



Entropy imposes a metric on the ensemble space

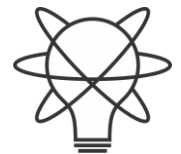
$$\|\delta \mathbf{e}\|_{\mathbf{e}} = \sqrt{8MS(\mathbf{e}, \mathbf{e} + \delta \mathbf{e})}$$

$$g_{\mathbf{e}}(\delta \mathbf{e}_1, \delta \mathbf{e}_2) = \frac{1}{2} \left(\|\delta \mathbf{e}_1 + \delta \mathbf{e}_2\|_{\mathbf{e}}^2 - \|\delta \mathbf{e}_1\|_{\mathbf{e}}^2 - \|\delta \mathbf{e}_2\|_{\mathbf{e}}^2 \right)$$

$$\Rightarrow g_{\mathbf{e}}(\delta \mathbf{e}_1, \delta \mathbf{e}_2) = -\frac{\partial^2 S}{\partial \mathbf{e}^2}(\delta \mathbf{e}_1, \delta \mathbf{e}_2).$$

Entropy strict concavity means the Hessian is negative definite

Recovers Fisher-Rao information metric
(both classical and quantum - BKM)

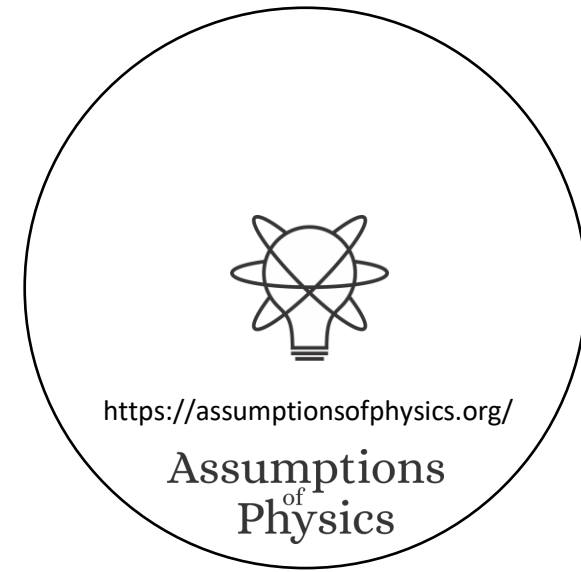
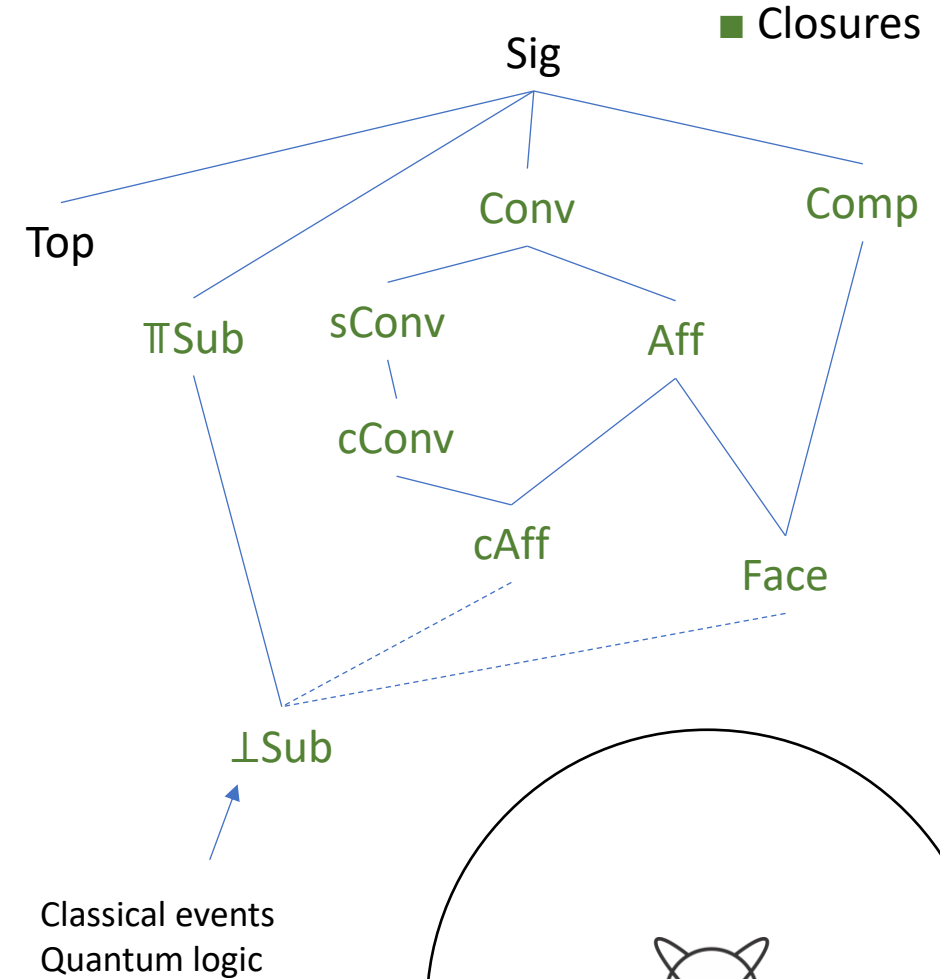


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Assumptions
of
Physics

Several useful lattices

- Experimental verifiability
 - Topology on the ensemble space (verifiable statements)
 - “This coin is fair within .1%” “The average energy is 5 ± 0.01 eV” ...
 - σ -algebra on the ensemble space (statements associated with a test)
 - “This coin is perfectly fair” “The average energy is a rational number in eV” ...
- Mixing
 - Convex sets (closed under finite mixtures)
 - σ -convex sets (closed under infinite mixtures – possibly convex integrals)
 - Closed convex sets (closed under mixtures and limits)
- “Unmixing”
 - Affine sets (closed under finite mixing/removal)
 - Closed affine sets (closed under mixing/removal and limits)
- Components
 - Componential sets (contain all components of their elements)
 - Facial sets (contain all components and mixtures)
- Subspaces (orthogonality space constructions)
 - Separate (no common component) subspaces (contain all separate to its separate: $V = (V^{\Pi})^{\Pi}$)
 - Orthogonal (mutually exclusive) subspaces (contain all orthogonal to its orthogonal: $V = (V^{\perp})^{\perp}$)



Generalization of Choquet theory

- How can we reformulate the idea of a simplex in a more general case?

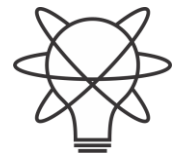
Corollary 1.319. *The π -subspaces are facial if and only if mixtures preserve separateness. That is, the following are equivalent:*

1. $\mathcal{E}$ is monodecomposable
2. mixtures preserve separateness in $\mathcal{E}$
3. every π -subspace is convex
4. every π -subspace is facial

If $e = pa_1 + \bar{p}a_2 = \lambda b_1 + \bar{\lambda}b_2$, then a_1 must have a common component with either b_1 or b_2

If $e \pi a$ and $e \pi b$, then $e \pi pa + \bar{p}b$

Preliminary result



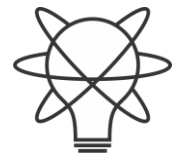
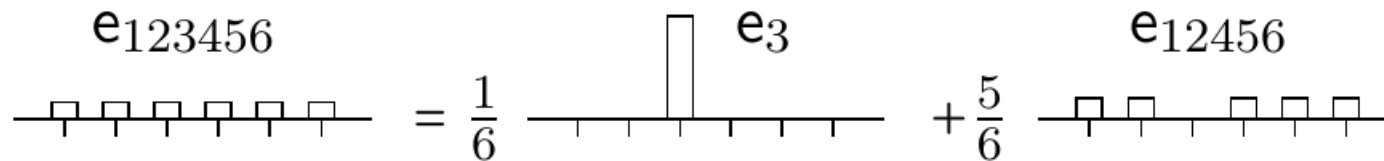
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Assumptions
of
Physics

Non-additive “probability measure”

Fraction: how much of one ensemble is made of another?

Definition 1.205. Let $\mathbf{e}, \mathbf{a} \in \mathcal{E}$ be two ensembles. The **fraction** of $\mathbf{a}$ with respect to $\mathbf{e}$ is the greatest mixing coefficient for which $\mathbf{e}$ can be expressed as a mixture of $\mathbf{a}$. That is, $\text{frac}_{\mathbf{e}}(\mathbf{a}) = \sup(\{p \in [0, 1] \mid \exists \mathbf{b} \in \mathcal{E} \text{ s.t. } \mathbf{e} = p\mathbf{a} + \bar{p}\mathbf{b}\})$.



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Non-additive “probability measure”

Fraction: how much of one ensemble can be constructed from others?

Definition 1.208. Let $e \in \mathcal{E}$ be an ensemble and $A \subseteq \mathcal{E}$ a subset. The **fraction capacity** of A with respect to e is the biggest fraction achievable with convex combinations of A . That is, $\text{fcap}_e(A) = \sup(\text{frac}_e(\text{hull}(A)) \cup \{0\})$.

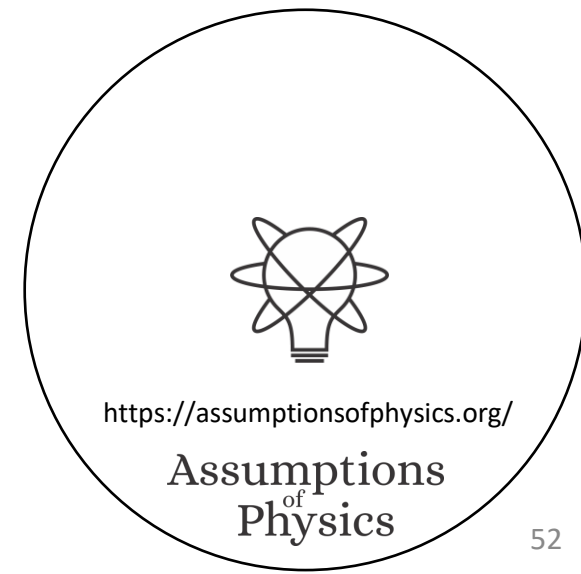
Recovers probability
over orthogonal subspaces

$$e_{123456} = \frac{1}{3} \left(\frac{1}{2}e_3 + \frac{1}{2}e_4 \right) + \frac{2}{3} e_{1256}$$

Proposition 1.211. The fraction capacity with respect to an ensemble is a set function that is

1. non-negative and unit bounded: $\text{fcap}_e(A) \in [0, 1]$
2. increasing: $A \subseteq B \implies \text{fcap}_e(A) \leq \text{fcap}_e(B)$
3. subadditive: $\text{fcap}_e(A \cup B) \leq \text{fcap}_e(A) + \text{fcap}_e(B)$
4. continuous from below: $\text{fcap}_e(\lim_{i \rightarrow \infty} A_i) = \lim_{i \rightarrow \infty} \text{fcap}_e(A_i)$ for any increasing sequence $\{A_i\}$

<https://arxiv.org/abs/2602.01470>



Non-additive state count

Count of states/spread of an ensemble must be given by the exponential of the entropy

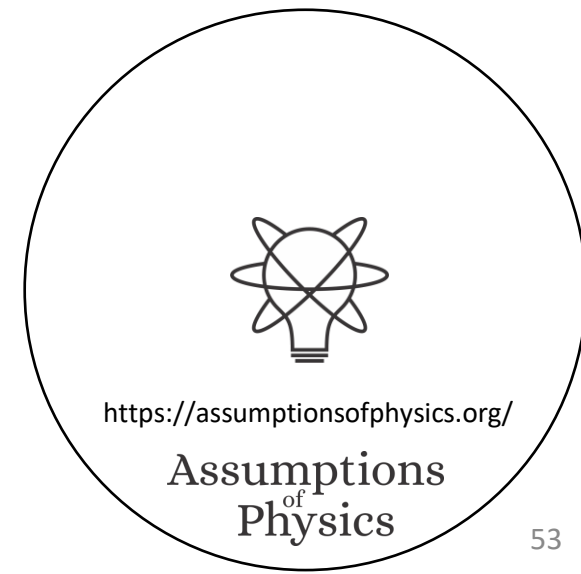
Proposition 1.194 (Uniqueness of exponential entropy). *Let $f : \mathbb{R} \rightarrow (0, +\infty)$ be a continuous increasing function such that $f(S(pa + \bar{p}b)) \leq f(S(a)) + f(S(b))$ for all $a, b \in \mathcal{E}$ and $p \in [0, 1]$, with equality for some $p \in (0, 1)$ whenever $a \perp b$. Then $f(S) = c \cdot 2^S$ for some $c > 0$.*

Definition 1.197. *Let $A \subseteq \mathcal{E}$ be a subset of an ensemble space. The **state capacity** of A is defined as $\text{scap}(A) = \sup(2^{S(\text{hull}(A))} \cup \{0\})$.*

Recovers $S(\rho_U) = \log \mu(U)$ in the classical case and the dimensionality of Hilbert subspaces in quantum

Proposition 1.200. *The state capacity is a set function that is*

1. *non-negative:* $\text{scap}(A) \in [0, +\infty]$
2. *increasing:* $A \subseteq B \implies \text{scap}(A) \leq \text{scap}(B)$
3. *subadditive:* $\text{scap}(A \cup B) \leq \text{scap}(A) + \text{scap}(B)$
4. *additive over orthogonal sets:* $A \perp B \implies \text{scap}(A \cup B) = \text{scap}(A) + \text{scap}(B)$
5. *continuous from below.*



The problem with counting on the continuum

We'd like:

1. Every state is a single case (i.e. $\mu(\{\psi\}) = 1$)
2. Finite continuous range carries finite information (i.e. $\mu(U) < \infty$)
3. Count is additive for disjoint sets (i.e. $\mu(\cup U_i) = \sum \mu(U_i)$)

Incompatible!

Pick two!

Discard 1 $\Rightarrow$ Lebesgue measure

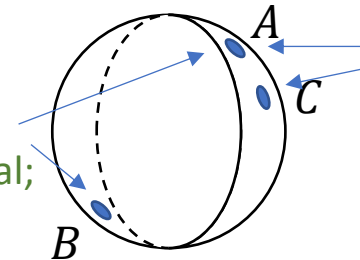
Discard 2 $\Rightarrow$ counting measure

Discard 3 $\Rightarrow$ "Quantum measure"

$$\mu(U) = 2^{\sup(s(\text{hull}(U)))}$$

Exponential of the maximum entropy reachable with convex combinations (statistical mixtures) of U (reduces to counting/Liouville measure)

Orthogonal states: additive
different states all else equal;
mutually exclusive



$$\mu(\{A\}) = 2^0 = 1$$

$$\mu(\{A, B\}) = 2^1 = 2$$

$$\mu(\{A, C\}) < 2 = \mu(\{A\}) + \mu(\{C\})$$

Non-orthogonal states: different states but in different contexts; not mutually exclusive

sub-additive

Quantum mechanics $\Rightarrow$ lower bound
on #conf (entropy) on continuous DOF



JOHN
TEMPLETON
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Conjecture: quantum gravity $\Rightarrow$ lower bound on DOF count

Same problem!

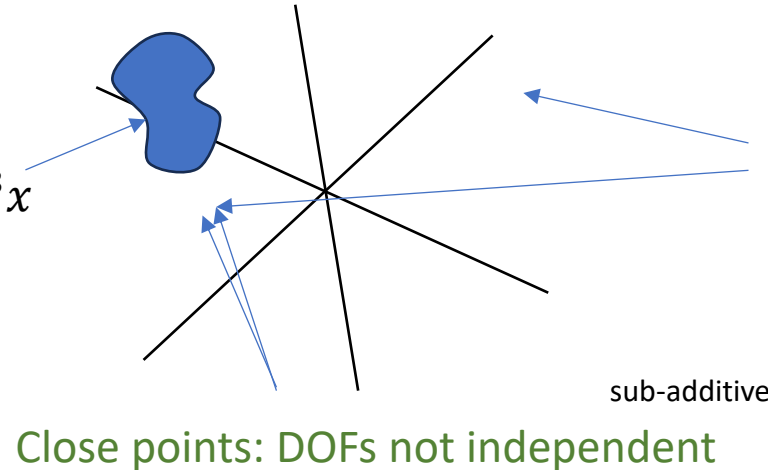
$$\frac{\#conf}{\#DOFs}$$

Lower bound
on this...

...requires a lower
bound on this

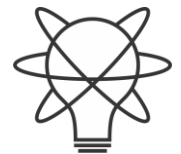
1. Every point is a single DOF (i.e. $\mu(\{x\}) = 1$)
2. Finite volume carries finitely many DOFs (i.e. $\mu(U) < \infty$)
3. Count is additive for disjoint regions (i.e. $\mu(\cup U_i) = \sum \mu(U_i)$)

$$\#DOFs \sim \text{Spatial volume } \int_U \sqrt{-g} d^3x$$



From QM: Lower bound on state count requires a severe revisitation of particle state space

Does lower bound on DOF count require an equally severe revisitation of space-time?

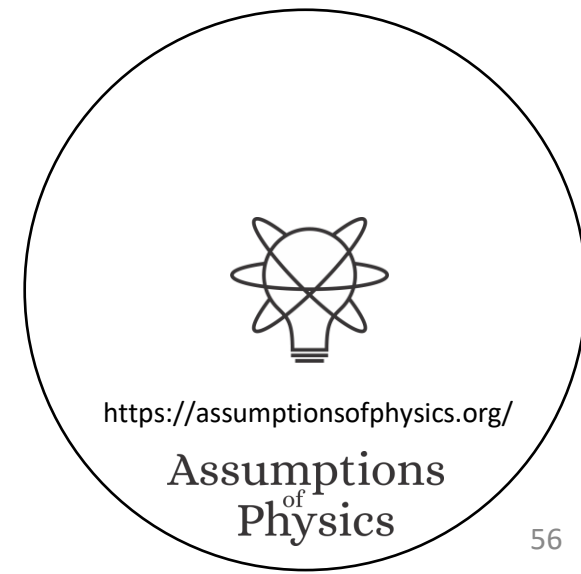


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Conclusion

- Ensembles are the natural primitives for physical theories
 - The math we use already implicitly assumes that states (even pure ones) are ensembles
 - Experimental reproducibility implies that ensembles are the topic of scientific laws
- A general theory of ensembles is therefore crucial to the foundation of physics
- There are four fundamental requirements
 - Experimental verifiability $\Rightarrow$ topology and σ -algebras
 - Statistical mixing $\Rightarrow$ convex structure
 - Ensemble variability $\Rightarrow$ entropic structure $\Rightarrow$ geometry
 - Ensemble as equilibria $\Rightarrow$ symmetries, Lie algebras
- Many open problems in math, physics and philosophy of science find a natural setting in this theory
- Lots of interesting work to be done!





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